

FRED TALKS

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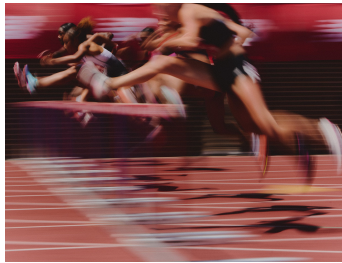
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Elite Young Runners Are Becoming Freakishly Fast. Welcome to ‘Trackflation’

BRENDAN I. KOERNER · 09 AUG 2026 · [SOURCE](#)

Four-minute miles are almost routine. Middle schoolers are beating college runners. Inside the youth track boom that’s turning kids and teens into speed machines.



PHOTOGRAPH: Elijah Agurs

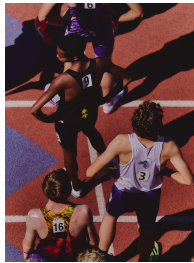
A couple of years ago, my teenage son decided to spend the long New York winter running indoor track. An accomplished soccer player, he joined the team with the modest goal of staying fit during his main sport’s offseason, but before long his competitive streak kicked in. He trained like a demon for two months straight, brutalizing his legs and lungs with high-intensity [workouts](#). The suffering paid off: His time in the 1,600-meter run (just shy of a mile) dropped so much that he earned a spot in our sectional championship, competing against runners from three nearby counties.

My son had the race of his life at sectionals, shaving several seconds off his personal best. He also got absolutely smoked. The top three boys all finished in well under 4 minutes and 30 seconds. The next six broke 4:38. By the time my son collapsed across the finish line in second-to-last place, the winner was deep into his post-race stretching routine.

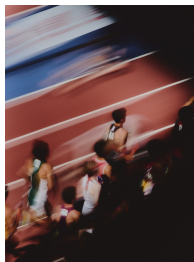
Had teenage [runners](#) at mid-tier meets always been that absurdly fast? Based on my limited knowledge of track, I suspected not, so I looked up the 1,600-meter results from the 2015 sectionals. Back then, only three boys broke 4:40, and my son would have finished in the middle of the pack. I was now curious whether kids beyond my provincial corner of New York were getting a lot faster too.

I followed a few track social media accounts, and soon the algorithms deluged me with highlights of young runners obliterating records on what seemed like a daily basis. A 12th-grade girl from Ohio ran an indoor 2-miler in an otherworldly 9:37.15, breaking a national high school record that had stood since 2013. A 10th-grade boy from Texas ran an outdoor 100-meter dash in 9.92 seconds, just a tenth of a second off the collegiate record. Crunch the data on any race in virtually any school-age meet and you'll find that the top and median times have plunged since the late 2010s, often by several seconds.

My favorite videos were made by a former University of Pittsburgh hurdler named Shyheim Wright, a master at channeling the bewilderment of track fans. "Pack it up, pack it up right now!" he marveled over footage of a ninth-grade girl from Georgia running 100 meters in 11.01 seconds, a time that made her faster than all but the very best college sprinters. "I'm so glad I don't run no more, I'm so glad. The times now is getting ridiculous."



PHOTOGRAPH: Elijah Agurs



PHOTOGRAPH: Elijah Agurs

The track world, I discovered, has coined a name for this much-discussed phenomenon of kids getting ultra-fast: *trackflation*. In June 2025 at the New Balance Nationals, an annual championship that draws around 6,000 top youth athletes, more than 100 kids beat previous meet records. "There might now be a hundred kids that run under 4:10 in the full mile," says Will Rodgers, who ran track at the University of Alabama in Huntsville in the early 2010s and now co-owns RunningLane, a coaching and events business. "Back when we were competing in high school, it might've been like 20 at max."

I had to know why kids today are essentially breaking track. And that meant trying to understand the dramatic shift happening at the sport's roots.

In early May, I drove down to Baltimore to attend the East Coast International Showcase, a two-day outdoor track meet sponsored by the shoe company Puma. One of its main draws was an 800-meter race featuring two fixtures of the elite youth circuit: Henry Bartle of Carmel, California, and Gunner Hammett of Williamsburg, Virginia.

As these two stars warmed up alongside Morgan State University's faded blue track, I spotted Hammett's dad wearing a custom hoodie that listed his son's credentials: "2 Time World Record Holder 800m, 18 Time All-American, 5-Time National Record Holder."

Those achievements were all the more impressive given Hammett's tender age: He was 13, just a bit younger than the 14-year-old Bartle. Both boys were still weeks away from their eighth-grade graduations.

The thin-limbed, bleached-blond Bartle ran the two-lap race in a personal best of 2:03.52, an outrageously good time. Three years ago that would have been enough to win the middle-school 800 meters at the New Balance Nationals. But he was nowhere close to Hammett, who coasted to a 1:57.75—a performance that put him on par with his home state's fastest high school runners.

"Guess I got smoked by you again," a genial Bartle told Hammett as they gasped for air past the finish line. Bartle has, unfortunately, grown accustomed to falling short. Seven weeks earlier, he and another kid both shattered California's age-group record for the indoor mile, but Bartle still finished six seconds off the podium. In the one-line caption for his Instagram post about that race, Bartle captured the essence of trackflation: "Living in a time where a state record still doesn't medal at Nationals."

Once Bartle had recouped enough oxygen to function, he gave me a thoughtful breakdown of his recent progress. "Last year, even if I had somewhat of an injury, I was very stubborn and I trained through it," he told me. Now he's prioritizing recovery as part of his regimen, and he believes a sub-two-minute 800 and a sub-4:30 mile are well within his grasp.

Until seven or eight years ago, few eighth-grade runners operated with Bartle's degree of maturity and determination. There were no middle-school divisions at the big national meets, and 13- and 14-year-olds rarely put serious weekly mileage on their legs. But perhaps inevitably, the youth sports boom expanded to include track.

As millions of exhausted parents can attest, sports such as soccer and baseball have become dauntingly competitive over the past generation. Even for third and fourth graders, there's a constant scramble for spots on prestigious travel teams, followed by nerve-racking fights for playing time. Some families decided they could avoid that madness by steering their kids toward track, which had relatively few participants at its entry-level ranks.

Grade schoolers began flocking to local track clubs. Dennis Robinson, head coach of the Zephyrs Track Club in Chicago, told me that for most of the 2010s he only managed to sign up high schoolers. "We were concerned about the future of our club," he said. "Now we're starting to see more and more younger athletes. And when I say young, I mean we've had to turn away athletes who are 4 years old." Another club I spoke with, the Youth Track Foundation in Portland, Oregon, signed up 60 kids in its first two years of existence and has a lengthy waiting list.

These clubs have also benefited from track's growing reputation as an antidote to the politics and favoritism that plague less objective sports. "I have so many parents that join our club, and they're just like, 'This is so great, because we had this terrible experience with the baseball team because this parent was coaching and he never let my kid pitch,'" says Brad Peterson, a cofounder of HP Distance, a track club in Indiana. "But in track there's no argument." The stopwatch determines everything.

The rise of track clubs was apparent on the second day of the East Coast International Showcase, when I watched wave after wave of kids aged 5 to 9 compete in the 100 meters. The many standouts ran times that would be dazzling for boys or girls several years older, and I took note of their efficient mechanics—evidence, it seemed to me, that they're already being coached to develop good habits.

When talented elementary-age kids from the club system move into their tweens and early teens, they're primed to start the grueling training that used to be reserved for high schoolers. "I'm asking middle-school girls to do eight 400s in 75 seconds, so at a 5-minute-mile pace," says Peterson. "Whereas it used to be the middle-school program was, 'Hey, let's jog 20 minutes and let's go play Capture the Flag,' right?"

Kids are increasingly doing that training in so-called "super shoes" such as the Nike Vaporfly 4 or New Balance FuelCell SuperComp Elite v5, both of which now often sell for well under \$200. These shoes' carbon plates and ultralight foams help return kinetic energy to the wearer's body with every step. In races, super shoes have been proven to boost performance by 2 to 3 percent, which translates to roughly a second per 400-meter lap.

Geoff Burns, a sports physiologist with the US Olympic & Paralympic Committee, believes the shoes' energy-return component decreases the traumatic stress endured by young bones and muscles. "If you're able to do that same work but not have that mechanical collateral damage or nearly as much of it, then you're just not building up as much scar tissue that could potentially inhibit performance," he says. Instead of diverting lots of cellular energy to the healing process after every workout, the body can dedicate more resources to building strength.



PHOTOGRAPH: Elijah Agurs

As middle schoolers have gotten faster, big-name meets have made room for them. New Balance Nationals added its middle-school division in 2023, and other prominent track events followed suit. Athletes like Bartle now spend a significant chunk of every winter and spring traveling across the country.

The results make track old-timers shake their heads with wonder. In April, for example, an eighth-grade boy shattered the national middle-school record for the mile (he ran it sub-4:18). Just two weeks later, an eighth-grade girl ran the 400 meters faster than any US high schooler had at that point in the year (she clocked 51.58).

It's not just the superstars who are getting faster. The runners in the pack behind them are picking up the pace, too, and they're transforming what's considered a decent time. "I think I have 10 middle-school girls who run between a 4:56 and a 5:16 mile," says Peterson. "Ten years ago, you'd be like, 'That's insane.' Now it's nothing." When these kids head to high school, they have the strength and experience to flourish right away. But given the sky-high expectations they face, the intensity can come at a cost.

There has long been a vast discrepancy between the number of high schoolers who run track and the money their programs receive. Outdoor track and field boasts more high school athletes than any other sport, with over 1.1 million participants—a figure likely inflated by the fact that teams tend to make few if any cuts. (The second-most populous co-ed sport, basketball, has fewer than 900,000 players.) Many athletic departments treat track like an afterthought, undeserving of the resources needed to hire coaches with real expertise.

“When I was coming through, my coach was a retired Marine who had never run a step in his life,” says Caryn Gehrke, an all-Ivy League runner at Columbia University in the early 2000s who now coaches in Ann Arbor, Michigan. “His entire philosophy was the more you hurt in practice, the more you can hurt in a race,” which is possibly the most counterproductive advice a runner can hear.

Today, a high school runner can scrutinize workout videos posted by pros or simply ask ChatGPT “How do I run a sub-2:00 800?” and develop a more effective training plan than anything their forebears were using in 2016. But for runners with any financial means, hiring a private coach for one-on-one guidance has become commonplace. It’s easy to find former collegiate runners on Instagram who are eager to capitalize on their years of geeking out on training methods. These coaches have popularized techniques such as heat acclimatization, in which athletes overexert themselves in sweltering conditions—say, by running on a treadmill next to a space heater—so that their bodies will produce extra plasma. (This is sometimes referred to as “legal blood doping.”)

They’re also a big reason why high school kids are dabbling in double-threshold training, which was pioneered by Norwegian pros. This approach involves doing a hard morning workout, then analyzing a drop of the athlete’s blood to see how much lactic acid their muscles produced. The runner then adjusts their pace up or down during a killer second workout of the day.



PHOTOGRAPH: Elijah Agurs



PHOTOGRAPH: Elijah Agurs

Private coaches also preach about proper nutrition and supplements. Survey the trash cans at any meet and you'll see empty tubs of Maurten Bicarb System, a sodium bicarbonate powder that runners consume en masse before races. The substance suppresses the development of lactic acid in muscles, allowing runners to stave off fatigue for a tad longer than normal. According to a 2021 meta survey, bicarb may boost performance by 1.5 percent, or around three seconds per mile.

There's plenty of chatter that some teen athletes are emulating the pros by ingesting banned substances—a cheat that's almost impossible to detect, since there's virtually no drug testing at the high school level. “I’m not saying I’m 100 percent on this, but I can pick out the athletes who are on it,” says Timothy Goldsack, a cofounder of the coaching service RunCCG, who cites human growth hormone and testosterone-replacement therapy as two popular performance enhancers. He blames familial pressure on kids to use track as a ticket to college: “Parents are crazy, you know that.”

To impress college scouts, high schoolers must travel to the national meets that are known for producing the most trackflationary times. These meets tend to have bouncy polyurethane tracks, favorable weather, or pace lights that keep runners apprised of the tempos needed to break records. A few well-known teen runners receive airfare and hotel rooms for gracing these meets with their presence, but most competitors spend small fortunes to attend. “If a family has the money to invest in that and it costs them \$2,000 to fly a kid across the country to take 15 seconds off their time, and that all of a sudden opens the door to another tier of scholarship or whatever, that’s a pretty good investment,” says Geoff Burns of the US Olympic & Paralympic Committee.

In mid-April, I spoke to Chris Catton, the other cofounder of RunCCG, as he prepared to coach several of his clients through a notoriously fast meet, the Asics Carolina Distance Carnival in North Carolina. “I have people who have flown here from Montana, North Dakota, Oklahoma, Rhode Island, New Jersey,” he told me by phone from his Airbnb in Charlotte. He was excited because the racing would take place at night, when times are fastest, and runners would be grouped together in heats according to their personal bests—a reliable way to get kids to push each other in a tight pack.

Two days later, Catton texted me a heads-up to check out the Carnival's results. They were staggering: In the high school boys' mile, the top 14 finishers all went sub-4:10. Five middle-school girls had broken the 5-minute barrier. Catton said that one of his clients, a ninth-grade girl who ran the 3,200 meters, had set an Oklahoma state record at the Carnival, yet only managed to come in 14th—half a minute behind the winner.

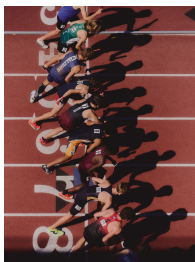
The social media content generated by these meets has turned a number of high schoolers into influencers. Some of them get sponsorship deals. Most receive perks rather than money—notably loads of free swag that they’re encouraged to flaunt online—though that may change as brands increasingly realize how much high school stars can shape the tastes of young consumers.

I watched one of those sought-after celebrities, 17-year-old Natalie Dumas, compete at the East Coast International Showcase. The New Jersey native had been one of the most recognizable faces of trackflation for nearly a year, ever since she’d won an incredible three races at the New Balance Nationals in June 2025. In Baltimore, she warmed up for a mere 20 minutes before demolishing the field in the 400-meter hurdles. She looked almost bored as she finished in 58.78; it wasn’t until I checked her Instagram later that I realized it was (at least at that moment) the fastest a high school girl had run the race this year.

When I caught up with Dumas right after the race, she implied that she would have run much faster if she wasn’t in such high demand. “I ran Wednesday, Thursday, Friday, and today’s Saturday, so this is my fourth meet this week,” she told me. “This is my sixth event in the past four days.”



PHOTOGRAPH: Elijah Agurs



PHOTOGRAPH: Elijah Agurs

For a brief moment in April, a stunning video was everywhere in my Instagram feed. It showed a 12-year-old girl from Utah running the mile in 4:40.98, a time that crushed the previous world record for her age group by nearly six seconds. She might have run even faster had she not elected to celebrate with the Scuba Dance a few steps short of the finish line.

The feat made Shyheim Wright, the content creator who specializes in highlighting the most ludicrous examples of trackflation, lose his mind. “Nah, bro, nah, bro, no way!” he began his Instagram post. “Bro, nobody was even in the camera with her, bro. There was other people in the race. I didn’t see nobody.” (“She was in my heat,” wrote one commenter, a seventh grader from North Carolina who ran her personal best in the race. “I didn’t see her once, she was so far ahead.”)

To many longtime track observers, however, the video was a cause for alarm. Multiple coaches and researchers told me this particular world record crystallized their fears for the safety of young runners who feel compelled to keep topping their peers. “When you see these amazing performances, it’s also just the tip of the spear of a lot of people also falling apart,” says Douglas Casa, a kinesiology professor at the University of Connecticut who studies elite athletes. “I’m not excited about all this progress. It just means that they’re working so much harder at a younger age.” He’s especially worried about young girls who specialize in middle- to long-distance events, since running the mileage required to excel in these races can delay puberty, leading to long-term issues such as lower bone density.

Burnout has always been a major problem in track, a sport anchored in athletes’ ability to sop up unnatural amounts of punishment. “I’ve spent my whole life running a race fueled by fear, never-ending expectations, and external validation,” wrote Stanford University 800-meter specialist Roisin Willis in a 2025 Instagram post about overcoming her mental health struggles; around the same time, Chloe Foerster, an All American middle-distance runner at the University of Washington, went public about her serious anxiety, which often caused her to vomit before races.

One 2022 study looked at the careers of 67,600 top-ranked teenage track-and-field athletes. Less than a quarter of 16- and 17-year-olds achieved comparable success past those ages, a transition rate the study authors described as “low.” Burns told me the results are even worse for kids who dominate at 16 or younger. “It’s a Faustian bargain,” he says. “Like, do you hyper-specialize and develop as a middle schooler and high schooler so you can get access and get scholarships and open all these doors, but it comes at the cost of actually being able to be good at it later?”

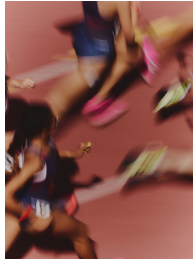
I didn't meet a single young runner who showed any trace of having pondered that question—naivete is one of the blessings of youth. They were instead wrapped up in envying the kids whose increasingly impressive victories get pumped into their phones every day: “When you see a girl run a faster time, all of them are trying to attack it, and they all have the mindset that they want to win,” said Dumas, the 17-year-old phenom.

Whenever it seems like times are about to plateau in a certain race, another 14- or 15-year-old emerges to set a new standard and stoke the competition. (That Utah girl who set the internet alight with a 4:40 mile in April? She beat her own record by two seconds in June.) Jesse Williams, a meet organizer in Seattle, used to give coaches a rough estimate of what times would earn their athletes a spot at the Brooks PR, an elite invitational that he helps oversee. He had to stop—records are falling too fast for Williams to keep up.

All that progress, of course, is built atop a foundation of pain, and I had to ask the kids why they love a sport that primarily tests their ability to endure agony. The answer that stuck with me most came from Dane Szczepkowski, an 18-year-old sprinter from SPIRE Academy in Geneva, Ohio. I caught him during a rough moment in Baltimore, mere minutes after he'd finished 39th in the 400 meters; he was clearly struggling as he tried to stretch away the damage he'd done to his legs. When I asked him why he'd become so devoted to running, he thought for a beat before offering an emphatic answer. “It's hard,” he said. “It's hard.”



Elite Young Runners Are Becoming Freakishly Fast.
Welcome to 'Trackflation



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I'd heard that sentiment from a lot of runners. They had chosen track specifically because it's an unglamorous grind—a lonely contest between them and a pitiless clock. The joy is in learning to appreciate the cause-and-effect relationship between arduous labor and incremental progress, even when that progress delivers a middle-of-the-pack result. Sophia Belcher, a high school junior from the Dallas area who won a Texas state title in the 800 meters this May, told me how the never-ending demands of track have been her rock through the most difficult moments of adolescence: “In a world where I think a lot of things are changing and not constant, I kind of had running that I could focus on, that would always be there.”

There's obviously nothing new about unmoored teens searching for structure, but track's rise does feel connected to the unique tumult of the past few years. Kids are understandably reeling from the echoes of the pandemic, social media, AI. Track just might be the ideal sport for those who tend to turn inward when looking to build self-belief. Amid the acceleration of everything, there's clarity to be found in trying to get a tenth of a second faster.

My son was good-natured about his thumping loss at the 2025 sectionals, but that race turned out to be the apex of his running career. He ran winter track again as a senior but his heart wasn't in it—he grew averse to the harsh training for the 1,600 and dropped down to shorter races, where faster sprinters dominated. A second trip to sectionals wasn't in the cards, which didn't bother him; by season's end, he'd become more interested in trying to learn how to kick field goals so he can walk on to his college's football team in the fall.

But our family's journey through track seems far from over. My 13-year-old daughter joined her middle-school team this spring, and after some trial and error, she settled on the 800 as her primary race. She gutted out some decent times at a few meets, and her coach rewarded her with a spot in a post-season meet held in a nearby college town.

She was nervous for days beforehand, and I tried to assure her that just showing up and running would be a victory—no one expects her to be the next Natalie Dumas. On the afternoon of the meet, though, she called me to report that things had gone much better than expected. Not only

had she conquered her butterflies, she'd also won her heat. She'd looked over her shoulder with about 50 meters to go and realized that no one was even close to overtaking her. "I pulverized those other girls," she said with evident glee.

She's still light-years away from a shout-out in a Shyheim Wright video. But in the era of trackflation, all things seem possible.

The images for this story were photographed at the New Balance Nationals Outdoor on June 21, 2026, at Franklin Field in Philadelphia.

The new rules of context engineering for Claude 5 generation models

CLAUDE.COM · 09 AUG 2026 · [SOURCE](#)

Then and now

There were a number of previous context engineering best practices that had become myths. Including:.

Then: Give Claude rules

Now: Let Claude use judgement

When we first rolled out Claude Code, we needed to be sure that Claude avoided worst case scenarios, such as deleting files. This meant we would give particularly strong guidance that might not always be true, For example, in the system prompt we used to say:

In code: default to writing no comments. Never write multi-paragraph docstrings or multi-line comment blocks — one short line max. Don't create planning, decision, or analysis documents unless the user asks for them — work from conversation context, not intermediate files.

But for a certain subset of prompts, this guidance would be wrong. In the case of documentation, the user may have their own preferences, or specific parts of very complex code might need multi-line comment blocks.

Still, without these guardrails for older models, the comments Claude wrote would be incorrect in many cases and we had to accept this tradeoff. But newer models have better judgement and can handle these decisions well without explicit rules.

In the new system prompt we say: *Write code that reads like the surrounding code: match its comment density, naming, and idiom.*

Then: Give Claude examples

Now: Design interfaces

The number one rule for tool usage was to give Claude examples on how to use them. With our newest models, we've found that giving examples actually constrains them to a certain exploration space.

Instead of using examples, think more about the design of your tools, scripts and files- what parameters does Claude have and how can they be more expressive?

For example, in the Todo tool example, just listing status as an enumeration between pending, in_progress, and completed, hints to Claude about how to use it. The instruction on keeping one item in_in_progress helps define our requested behavior.

Then: Put it all upfront

Now: Use progressive disclosure

Because Claude Code was focused on coding, our system prompt included detailed information on how to do code review and verification. These were not always needed, but when they were, it was crucial information.

Since then, Claude Code has gotten very competent at using progressive disclosure- loading the right context at the right times. For example, we moved verification and code review into their own skills that Claude Code could selectively call.

But progressive disclosure is not just for skills, we also use it for tools. Some of our tools are 'deferred loading,' which means the agent must search for their full definitions using ToolSearch before using them. This allows us to have more tools (such as our Task tools) that don't take up context until they're needed.

The same can be applied to your own CLAUDE.md and Skill.md files. A common myth is that you want to make these a central repository for every known practice that you *might* run into, because Claude would not find it otherwise. Instead, consider having a tree of files that can be loaded at the right time.

Then: Repeat yourself

Now: Simple tool descriptions

Earlier Claude models could sometimes need repeated instructions or be more likely to listen to instructions at the end of their context window than at the start. This meant our system prompt would sometimes have references to tools in the main system prompt as well as instructions in the tool description.

We found we could delete these repeat examples and put instructions on how to use tools in the tool descriptions rather than the system prompt.

Then: Memory in CLAUDE.md files

Now: Auto-memory

We used to encourage users to save things to Claude's memory, by using the # hotkey to write to their CLAUDE.md automatically. Instead, Claude now automatically saves memories that are relevant to the work and to you.

Then: Simple specs

Now: Rich references

In plan mode, Claude Code has heavily relied on markdown files with plans. Storing these files as plans helped Claude refer to them when needed. Another similar best practice was to store specs in the codebase for Claude to refer to while working across longer projects.

But we've found that Claude can handle increasingly more complicated references. Instead of simple markdown files, Claude can reference HTML artifacts created by our new artifacts feature.

You may also give Claude references in the form of code. A spec may also be a detailed test suite, or a function in a different codebase that Claude might port.

Rubrics are another form of references. Rubrics allow Claude to try and verify your taste in a particular field (e.g. what does a good API design look like) by using [dynamic workflows](#) and spinning up verifier agents with those rubrics.

Applying this to your context

Pulling this all together, what does this look like when you assemble your context?

System Prompt

A system prompt is heavily tied to the product context. It tells Claude what product it's operating in and what it's doing. For Claude Code, you will likely never modify this, but if you are building your own agent harness, this is where you should spend a lot of time.

CLAUDE.md

Keep your CLAUDE.md lightweight and briefly describe what your repo is for, but spend most of the tokens on gotchas inside of the codebase. For example, you may organize your code to keep types in one monolithic file and nowhere else. Avoid stating 'the obvious' things Claude should know by looking at your file system or your repo.

Use progressive disclosure heavily, for example if you have several unique instructions on how to verify your work, create a verification skill and reference it from your CLAUDE.md.

Skills

Think of skills as lightweight guides to let Claude find information when needed. Avoid making them overconstrained, except in highly important areas.

For long skills, try and use progressive disclosure as much as possible- divide it into many files and split them out.

It's best when skills encode particular opinions, knowledge, or best practices that are particular to you, your team, or product.

References

You can @ mention files to include them as references. References allow Claude to refer to in-depth information about the current plan.

This might be in specs files, mockups, or even entire codebases. Generally you should prefer files that are in code as it provides clear, high-fidelity instructions to Claude in a language it knows very well. For example, a HTML mockup of a design will generally produce better results than a description of the design or a screenshot.

Try simplifying

Across your system prompt, skills, and CLAUDE.md files, you may need to simplify just like we did. We rolled out a new command called `claude doctor`, which will help you do this automatically as well. For more details on prompting more advanced models specifically, check out our [Fable field guide](#).

This article was written by Thariq Shihipar, member of technical staff, Anthropic.

The Reverse Information Paradox

SN SCRATCHPAD · 05 AUG 2026 · [SOURCE](#)

Notes on advances in technology and real-world impact

In the age of intelligence, how should firms protect their core IP?

Nobel Prize winning economist Kenneth Arrow [famously described](#) a paradox in the market for information. “Its value for the purchaser is not known until he has the information, but then he has in effect acquired it without cost.” In Arrow’s “Information Paradox,” the seller risks giving away knowledge in order to sell it.

AI creates the reverse problem. In the AI age, the buyer risks giving away knowledge, just in order to use what they bought.

You essentially pay for intelligence twice, once with money, and again with something even more valuable: the proprietary knowledge you must reveal to make that intelligence useful. The better you want the model to perform, the more of that knowledge you have to feed it!

Over time, the information asymmetry becomes increasingly skewed. The seller learns more and more about you as you use what you purchased, while you learn very little about what the seller is learning in return.

That is what I think of as the Reverse Information Paradox.

Patents solve one aspect of Arrow's paradox. They let an inventor disclose an idea without simply giving it away. The Reverse Information Paradox needs its own equivalent.

This requires more than data protection. Models learn from “exhaust,” the prompts people write, the tools agents use, and especially the corrections people make when the model is wrong. Every correction is distilled into institutional know-how. It's the kind of knowledge a competitor could never buy, and the kind that leaks almost imperceptibly: trace by trace, correction by correction, eval by eval.

In consuming intelligence, you are creating intelligence. And what you create should belong to you. This is your particular intelligence, in Hayek's sense: the knowledge of time, place, and circumstance that no one else can hold. It knows what you think, what you value, and how you measure success.

While the great innovation that comes from model providers having fair use rights to train models on public data is needed, I find it ironic that the status quo is to then turn around and impose restrictive terms on distillation, and to reserve the right to learn from customer usage and interaction data. If learning flows in only one direction, economic value converges toward the owners of the learning infrastructure rather than the creators of the knowledge itself. Therefore, it's imperative that we distribute the learning infrastructure to every firm so that they can control their own learning loop.

As Alex Karp put it: “What the technical customers want is control over their compute, their models, their data stack, and their alpha. They want to know they own the means of production, and it's not being transferred to someone else.” The current regime does precisely the transfer Karp and companies fear.

That is why enterprises need a real trust boundary for their human capital and token capital to compound. It is where an organization's data, traces, evals, adapted weights, and memory accumulate and improve together. And it is a hard boundary across which nothing crosses, not

even the intelligence exhaust, without consent. Enterprises will demand the rights to use model outputs to fine tune and/or train their own models. I think of this as every firm's right to align models to their enterprise accountability obligations.

In the cloud era, enterprises accumulated data. In the AI era, they accumulate learning. The trust boundary must evolve accordingly, from protecting information to protecting the mechanisms through which organizations learn, adapt, and compound intelligence. There are a few things every enterprise must do to ensure this:

Control: Create your private evals, because evals define what “good” looks like inside the organization. Also, retain ownership of your organization's memory, traces, feedbacks, decisions, and institutional context, and ability to use outputs of models from your own tasks and queries.

Capability: Build your own proprietary learning environments within the tenant boundary to train or tune models, where models learn against real workflows without exposing the company's knowledge.

Choice: Ensure the orchestration layer is decoupled from any single model. Ask yourself: If any one model you are using is taken away, do you still have the ability to operate and optimize for your evals using other models? Does your company “veteran” capability remain with you even if a given “generalist” model is taken away?

Cost: By decoupling the orchestration layer, you are also able to bring together context, models, and tasks in the most efficient and cost-effective way without sacrificing quality.

Compound: Bring these four together and you create your own continuous learning loop (i.e. hill climbing machine) that will allow your AI investments to compound the value of your firm.

In other words, a company should be able to use a model without giving up the knowledge that makes it unique. That is the reverse information paradox we need to confront.